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WHITE PAPER · 2026

# Practice Before Patients

What the research on Virteasy haptic simulation means for learning pathways and operational productivity in dental schools

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Saratoga · Exclusive distributor of Virteasy Dental, USA and Canada

EXECUTIVE SUMMARY

# Simulation as a practice layer, not a replacement

Virtual haptic simulation is not a replacement for the phantom lab or the clinic. Across 18 studies and two reviews on Virteasy [1–21], it works best as a **pre-clinical practice layer**: students repeat procedures at any hour, with objective data on every attempt, before they touch a phantom head or a patient.

### Learning pathway

Randomized and comparative studies in implantology, endodontics and restorative dentistry report faster or better early skill, and lower stress and anxiety, when simulator practice comes first. [1–6]

### Productivity

Objective scoring is documented. Gains in instructor time, consumables and lab scheduling are plausible but **not yet measured** in controlled studies. A pilot can measure them. [2]

### Honest limits

Not every result is favorable. A headset lowered performance, and a plate-based model beat the simulator in one trial. No study reports cost or long-term retention. [8, 10]

**232** simulators in 70 institutions

**28** countries with installations

**18** studies reviewed here

**2** systematic and scoping reviews

## What administrators can take away

1. Treat the simulator as an added practice step with entry and exit criteria for each skill, not as a substitute for phantom hours.
2. Prefer the headset-free set-up for complex tasks such as access cavity preparation.
3. Budget for faculty calibration and for a transition period back to conventional simulators.
4. Measure before scaling: use a 90-day pilot with KPIs for instructor hours, lab occupancy, consumables and time to competence.

Sources: numbered references in square brackets are listed on page 11. Installation figures: list supplied for this paper (page 4).

THE LEARNING PATHWAY

# Where simulation fits between theory and the clinic

A dental curriculum moves students from knowledge to supervised patient care. The research positions the simulator in the middle of that path, between theory and phantom work.

| | | |
|---|---|---|
| 1 | **Orientation**<br>Anatomy, instruments and ergonomics on screen. | Students and staff rate it useful (127 and 865 respondents) [16, 17] |
| 2 | **Fundamentals**<br>Repeated drills in implant, cavity and access preparation. | Better or faster early results (88 and 40 students) [2, 4] |
| 3 | **Feedback**<br>Objective scores and expert review of each attempt. | Agrees with teachers 71 to 78% of the time [12] |
| 4 | **Transfer**<br>Phantom-head work, with a planned transition period. | Haptic scores did not predict phantom results [9] |
| 5 | **Clinic**<br>Supervised patient care once the skill is shown. | Case reports only; no outcome trials [15] |

**Design principle.** Move students on by demonstrated competence, not hours logged. A 2025 Sheffield thesis [22] adds that breaking a task into parts can help, and that motor-skills research does not support replacing the phantom head with VR. A 2017 thesis from Nancy [23] adds that performance stalled or fell once the angulation aid was removed after the fourth attempt, so aids should be withdrawn gradually, with briefing and debriefing.

Sources: references 2, 4, 9, 12, 15, 16, 17, 22 and 23 (page 11).

ADOPTION

# A platform already used in dental schools worldwide

The installation list supplied for this paper records 232 simulators in 70 institutions and 28 countries between 2015 and 2025 (supplied list, not independently verified). Twenty-five sites are marked as classroom set-ups.




**Universities and schools with Virteasy installations**

Source: Virteasy installation list supplied for this paper, 2015 to 2025; see Methods on page 11. Institutions shown appear in that list.

THE EVIDENCE · PART 1

# Controlled studies: promising early skill, with caveats

Nine controlled or comparative studies. They differ in design and topic, so read them as a map of signals, not as pooled proof.

| STUDY | DESIGN | SAMPLE | SIGNAL | TAKEAWAY |
|---|---|---|---|---|
| **Joseph, 2014** [1] | Randomized; implant drilling, simulator vs lecture only vs expert dentists | 60 (20/20/20) | Positive | Simulator-trained students improved over 8 trials and moved toward expert level. Resin-model perforations 10% vs 20%, not formally tested. |
| **Vincent, 2020** [2] | Randomized; cavity preparation on Virteasy vs plastic teeth, final test on plastic | 88 first-year | Positive | Both groups improved and final results were similar. The authors cite savings in supervision time and materials. |
| **Vincent, 2022** [3] | Randomized; implant site with vs without virtual aids | 88 first-year | Positive | Students with aids got better results faster; repeated practice built skill without aids. |
| **Usta, 2025 (Int Endod J)** [4] | Randomized; VR first vs artificial teeth first | 40 | Positive | VR first gave higher dexterity and self-rated proficiency and lower anxiety. Stress fell in both groups. |
| **Usta, 2025 (J Endod)** [5] | RCT, 3 arms: simulator, mobile app, control | 30 | Positive | Reported better clinical performance, lower stress and higher confidence at first clinical procedures. |
| **Al Ali, 2024** [6] | Crossover; 3D vs 2D vision on the simulator | 24 (21 finished) | Positive | 3D vision improved accuracy ($p = 0.035$) and reduced over-cutting. Time did not differ. |
| **Bandiaky, 2026** [8] | Crossover; access cavity, with vs without VR headset | 90 | Caution | Better haptic parameters without the headset. Most students preferred the non-immersive set-up. |
| **Yildirim-Manav, 2026** [10] | Randomized, 3 groups: VR haptic, CaviPrep plate, control | 75 | Caution | The plate group did better on outline, retention form and width. Limited haptic realism restricts standalone use. |
| **San Diego, 2022** [11] | Two cluster-randomized trials; haptic vs phantom, caries removal | 264 | Mixed | Similar overall learning, but the haptic group scored lower on a harder task and on instrument holding. |

Signal: Positive = favors simulator practice; Mixed = benefits and limits; Caution = result limits use. The San Diego paper does not name its haptic system.
Sources: references 1 to 6, 8, 10 and 11 (page 11). Numbers in brackets identify each study.

THE EVIDENCE · PART 2

# Feedback, feasibility and what students say

Nine studies on assessment, feasibility and student perception. They show acceptance and workflow potential, and they also show where the experience falls short.

| STUDY | DESIGN | SAMPLE | SIGNAL | TAKEAWAY |
|---|---|---|---|---|
| **Dixon, 2020** [12] | Validity test; simulator feedback vs 12 teachers, 10 printed cavities | 12 teachers | Positive | Simulator and teachers agreed 71% (pooled) to 78% (modal) of the time. One exercise only. |
| **Carpegna, 2023** [14] | Feasibility; endodontic microsurgery built from patient scans | Not reported | Feasibility | Virtual patient cases give reviewable feedback and saved progress. |
| **Pasqualini, 2025** [15] | Case report; 10 virtual sessions, then supervised surgery | 1 resident | Feasibility | Resident reached target scores, then operated; the lesion healed completely at 1 year. One case only. |
| **Towers, 2021** [13] | Interviews; VR then 3D-printed patient-specific model | Not stated | Perception | Students felt more confident and prepared before treating their patient. |
| **Nassief, 2024** [7] | Qualitative; 2D vs 3D vision | Not stated | Perception | Students valued 3D and saw VR as an early adjunct to the phantom head. |
| **Fernandez-Sagredo, 2016** [16] | Cross-sectional survey, Chile | 127 | Perception | 94% positive for teaching manual skills, 91% for autonomous practice, 72% for assessment. |
| **ADEMA survey, Spain** [17] | Exploratory survey and interviews | 865 | Perception | 86.7% wanted this earlier in training. Authors are affiliated with the adopting school. |
| **Faid, 2026** [18] | Cross-sectional survey; pediatric pulpotomy | 115 | Mixed | 63.6% satisfied, but 52.7% found it difficult to use (ergonomics, tactile realism). |
| **Bandiaky, 2025 (J Dent Sci)** [9] | Prospective pilot; haptic exercises vs crowns on phantom | Not stated | Mixed | Haptic performance index did not correlate with conventional results; a transition period is needed. |

Sources: references 7, 9 and 12 to 18 (page 11). Numbers in brackets identify each study.

WHAT USERS REPORT

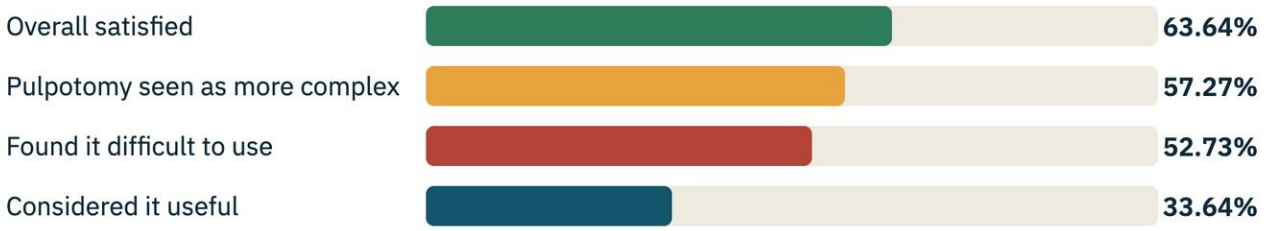
# Strong on teaching and practice, weaker on realism

Two surveys give the clearest view of how users judge the simulator. Teaching and self-practice score highest; realism and ease of use score lowest.

**Chile survey, 127 students, dentists and faculty: share positive** [16]




**Morocco survey, 115 students, pediatric dentistry: share of students** [18]




## +3D

Stereoscopic vision improved accuracy ($p = 0.035$) with no difference in task time. [6]

## -Headset

Performance was better without the VR headset in 90 third-year students. [8]

## 1 to 4

Studies agree that practice should be added to, not replace, phantom work. [2, 7, 9, 22]

Percentages as reported in each source. Different questions and samples mean figures are not comparable across surveys.

Sources: references 2, 6, 7, 8, 9, 16, 18 and 22 (page 11).

LIMITS AND GOOD PRACTICE

# What the evidence cannot yet say

A credible case for simulation names its limits. The reviews and trials point to six gaps that matter for a dental school budget.

### No cost data

No study reports cost per student, equipment lifetime or return on investment. A 2021 review says high initial cost is a main barrier [20]; the 2025 case report notes high room costs [15].

### No retention data

Skills are measured at the end of training, not months later in the clinic.

### No patient outcomes

Links between simulator scores and patient care are untested; only case reports exist [15].

### Realism and ergonomics

A 2019 scoping review found most haptic arms cannot resist rotation, and many set-ups lack a finger rest [19].

### Tutors still needed

The same review finds simulator feedback is not sufficient for unsupervised use [19].

### Developer links

Some research teams collaborate with the developer [19], and the core haptic algorithm comes from research that led to the developer, Didhaptic (Duriez 2013 [24, 25]). Independent replication would strengthen the case.

**DO**

- Add simulator practice before phantom work.
- Use headset-free mode for complex preparations. [8]
- Calibrate faculty on the scoring.
- Plan a transition period to the conventional unit. [9]
- Withdraw guidance aids gradually and add debriefing. [23]

**AVOID**

- Replacing phantom or clinical hours by default.
- Judging students on simulator scores alone.
- Buying seats before measuring local demand.
- Claiming savings before data exist.

Sources: references 8, 9, 15, 19, 20 and 23 to 25 (page 11).

OPERATIONS LENS

# Where productivity could come from

For business and clinic administrators, seven operational levers are worth testing. Only objective assessment data is directly documented.

| | | |
|---|---|---|
| **Assessment data** | Objective metrics and saved progress for every attempt | Documented |
| **Time to competence** | Vincent 2020 [2]: equal final results; a review [19–21] cites 22% shorter procedure time (543 to 424 s). 3D vision showed no time effect [6]. | Early signal |
| **Instructor time** | Vincent 2020 authors cite saved supervision and teaching time; it was not measured as an outcome in the abstract [2] | Early signal |
| **Consumables** | A 2021 review [20] argues phantom materials and waste are recurring costs; a 2017 thesis notes exercises can be repeated without extra cost (author statement, not measured) [23] | Opportunity |
| **Lab scheduling** | Practice available outside supervised lab slots | Opportunity |
| **Case build time** | A new patient-specific scenario took weeks to months to create (new avatar 2 to 4 months; a further case about 2 weeks) [13–15] | Documented |

## Pilot KPIs

- **Instructor hours per student per skill**
- **Lab occupancy and idle hours**
- **Consumables per student**
- **Time to competence**
- **Students needing remediation**

Sources: references 2, 6, 13 to 15, 19 to 21 and 23 (page 11). Pilot KPIs are a Saratoga proposal, not a finding.

NEXT STEP

# A 90-day pilot to measure what matters

A short, structured pilot turns assumptions into local numbers.

Days 1 to 15

### Baseline

Record current hours, lab use, consumables and skill scores.

Days 16 to 30

### Set-up

Install simulators, calibrate faculty, define entry and exit criteria.

Days 31 to 75

### Run

Run one cohort with simulator practice and one with the usual curriculum.

Days 76 to 90

### Review

Compare KPIs, collect student feedback, decide on scale-up.

ABOUT SARATOGA

Saratoga is the exclusive distributor of Virteasy Dental in the United States and Canada. Based in Pordenone, Italy, Saratoga also manufactures dental simulators and develops EasyTeachCloud, an e-learning platform that supports students' learning pathway. Its latest research project, ASISME (Advanced Simulation Student Monitoring Ecosystem), combines artificial intelligence, computer vision, blockchain and a digital twin to record how students interact with the simulators, assess their performance objectively and evaluate their working posture.

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Sources: the 90-day plan is a Saratoga proposal, not drawn from the cited studies. Company information supplied by Saratoga.

REFERENCES AND METHODS

# Sources and how this paper was built


1 Joseph D et al. Biomed Res Int 2014;413951.

2 Vincent M et al. J Dent Educ 2020;84(3):367-376.

3 Vincent M et al. J Dent Educ 2022 (doi 10.1002/jdd.12916).

4 Usta SN et al. Int Endod J 2025 (doi 10.1111/iej.14236).

5 Usta SN et al. J Endod 2025 (pii S0099239925007113).

6 Al Ali H et al. J Dent Educ 2024 (doi 10.1002/jdd.13630).

7 Nassief S et al. J Dent Educ 2024 (doi 10.1002/jdd.13682).

8 Bandiaky ON et al. Int Endod J 2026 (doi 10.1111/iej.14252).

9 Bandiaky ON et al. J Dent Sci 2025 (pii S1991790224003660).

10 Yildirim-Manav E et al. Eur J Dent Educ 2026 (doi 10.1111/eje.70201).

11 San Diego JP et al. Dent J 2022;10(11):198.

12 Dixon J et al. Eur J Dent Educ 2020 (doi 10.1111/eje.12581).

13 Towers A et al. Eur J Dent Educ 2021 (doi 10.1111/eje.12715).

14 Carpegna G et al. Eur J Dent Educ 2023 (doi 10.1111/eje.12946).

15 Pasqualini D et al. Int Endod J 2025 (doi 10.1111/iej.14239).

16 Fernandez-Sagredo M. ADEE 2016; Spanish journal version 2020.

17 ADEMA student survey. Medicina Balear 2022;37(3):142.

18 Faid K et al. Cross-sectional study, 2026, eu-opensci.org.

19 Towers A et al. Br Dent J 2019 (scoping review, 68 papers).

20 Imran E, Adanir N, Khurshid Z. Appl Sci 2021;11(21):10196.

21 Dzyuba N et al. Eur J Dent Educ 2022 (doi 10.1111/eje.12871; 14 RCTs, one on Virteasy).

22 Towers AC. Deconstructed learning in pre-clinical dental education. PhD thesis, Sheffield, 2025.

23 Joseph D. Impact de la simulation haptique dans l'enseignement en odontologie. Thesis, Université de Lorraine, 2017.

24 Syllebranque C, Duriez C. 6-DoF haptic rendering for dental implantology simulation. ISBMS 2010:139-149.

25 Duriez C. Real-time haptic simulation of medical procedures (HDR thesis), Université de Lille, 2013.


## Methods and notes

Sources come from the research list supplied for this paper and were read in full or in abstract; summaries are paraphrased from abstracts and open full texts. Vincent 2020 and 2022 are two separate trials; the 2022 paper is listed under Joseph in the source file. The 2019 scoping review declares a research collaboration with the Virteasy developer. The haptic algorithm behind Virteasy was co-invented by the author of ref 25, and the developer is Didhaptic (refs 24, 25). Saratoga distributes Virteasy Dental in the United States and Canada, and readers should weigh the evidence with that commercial interest in mind. Installation figures come from the supplied list and are not independently verified. Productivity levers are hypotheses to test in the pilot, not findings.

Numbers in square brackets throughout this paper refer to this list.